Tucker McElroy
U.S. Census Bureau
United States

Recursive algorithms, based upon the nested structure of Toeplitz covariance matrices arising from stationary processes, are presented for the efficient computation of multi-step ahead forecasts (with mean squared error) and signal extractions. The methods are developed for nonstationary vector time series, and are applied to high frequency retail and immigration data exhibiting trend dynamics and multiple types of seasonality.

Natural Language Processing for Predicting Everyday Behavior with and without Time and Duration Information

Rafal Rzepka
Hokkaido University
Japan

Kenji Araki
Hokkaido University
Japan

Forecasting is a complicated process which is impossible without sufficient and reliable data. Researchers and data specialists have been predicting sales, epidemics, stocks, elections and many other possible outcomes which could be useful to know beforehand. People do not need to forecast phenomena that are characterized by easily predictable consequences but artificial agents are bad at forecasting very obvious human's everyday behavior. This is because data on everyday life is rarely available, being meant for commercial use or due to privacy issues, which makes machine learning difficult or very narrow. If such data is collected and shared, usually only a few specific aspects of behavior are recorded together with time stamps. In our approach we combine text mining and natural language processing for discovering order of acts with sentiment analysis to predict usual outcomes of human's deeds. In our presentation we introduce results of our experiments (Rzepka 2015) with predicting consequences of ethical and unethical actions from Japanese blogs (5,5 billion word raw text corpus, almost 80% agreement with human subjects). Then we report results of experiments with Twitter data and discuss a possibility of natural language processing techniques to automatically recognize and represent time duration in machine-readable format (Krawczyk 2013), which could be a helpful addition for many types of forecasting relying only on time-stamps. By presenting our findings in predicting judicial verdicts, we also share our suggestions on how text-similarity algorithms could be utilized by forecasting community. Then we compare our approach with a method for extracting future reference sentences using semantic role labeling (Nakajima 2016). We conclude with a discussion on how forecasting methods can be useful for common sense knowledge acquisition and how machines could become able to automatically discover trends worth forecasting, for example by machine reading of scientific papers.

(Krawczyk 2013) Marek Krawczyk, Yuki Urabe, Rafal Rzepka and Kenji Araki, "A-dur: Action Duration Calculation System", SIG-LSE-B301-7, pp.47-54, 2013. (Rzepka 2015) Rafal Rzepka, Marek Krawczyk and Kenji Araki, "Replacing Sensors with Text Occurrences for Commonsense Knowledge Acquisition”, Proceedings of IJCAI 2015 Workshop on Cognitive Knowledge Acquisition and Applications (Cognitum 2015), Buenos Aires, Argentina, July 25-31, 2015. (Nakajima 2016) Yoko Nakajima, Michal Ptaszynski, Hirotoshi Honma, Fumito Masui, "Automatic extraction of future references from news using morphosemantic patterns with application to future trend prediction", AI Matters 2(4): 13-15, 2016.